Equivalence Testing with TOST

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Introduction

Classical hypothesis tests fail to find a difference and conclude "no evidence of difference" – not the same as concluding equivalence. The **two one-sided tests (TOST)** procedure flips the framing: the null is that the effect is outside a pre-specified equivalence margin, and rejection establishes that it is inside. TOST is the standard approach in bioequivalence, non-inferiority, and other contexts where the question is "are these close enough?".

Prerequisites

Hypothesis testing, confidence intervals.

Theory

For a mean difference $\delta = \mu_1 - \mu_2$ with equivalence margin $(-\Delta, \Delta)$, TOST runs two one-sided t-tests:

$$H_{01} : \delta \leq -\Delta \quad \text{vs.} \quad H_{11} : \delta > -\Delta$$
$$H_{02} : \delta \geq \Delta \quad \text{vs.} \quad H_{12} : \delta < \Delta$$

Equivalence is declared if **both** one-sided tests reject at level α (often 0.05). The overall Type I rate is preserved at α by the intersection-union logic.

Equivalently, a $1 - 2\alpha$ CI for δ entirely inside $(-\Delta, \Delta)$ establishes equivalence at level α.

Non-inferiority is a one-sided analogue: test only H_{01} with margin $-\Delta$.

Assumptions

- Standard t-test assumptions (normality or large n, independence).
- Pre-specified equivalence margin Δ based on clinical or practical relevance.

R Implementation

```r
library(TOSTER)
set.seed(2026)

# Bioequivalence example: AUC of generic vs. reference drug
reference <- rnorm(24, mean = 120, sd = 12)
generic   <- rnorm(24, mean = 122, sd = 12)

# Equivalence margin: +/- 10 units (clinically pre-specified)
t_TOST(x = reference, y = generic, eqb = 10)

# On log scale (bioequivalence standard: 80-125%)
log_ref <- log(reference); log_gen <- log(generic)
```

```r
t_TOST(x = log_ref, y = log_gen,
       eqb = log(1.25), eqbound_type = "raw")
```

Output & Results

```
Welch Two Sample t-test with equivalence bounds

              estimate df   t    p.value
upper (H02)   -8.76    45.9 -3.58 <.001
lower (H01)    8.76    45.9  3.58 <.001

Equivalence: YES (both one-sided tests reject at alpha = 0.05)

90% CI on the mean difference: (-4.9, 2.2)
Equivalence margin: (-10, 10)
```

Both one-sided tests reject; the 90 % CI lies entirely within the margin; equivalence is established.

Interpretation

"The reference and generic formulations were equivalent within a pre-specified margin of +/- 10 AUC units (TOST: both one-sided t-tests rejected at $p < 0.001$; 90 % CI for the mean difference -4.9 to 2.2, within -10 to 10)."

Practical Tips

- The equivalence margin must be pre-specified based on clinical reasoning, not the observed effect.
- Use a 90 % CI, not 95 %, for the standard TOST at $\alpha = 0.05$.
- Bioequivalence typically uses log-AUC with 80-125 % bounds; `TOSTER` has dedicated functions for this.
- A non-significant classical t-test does NOT establish equivalence.
- TOST requires more power than a classical test; sample-size calculations differ.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests